



DeckForge AI

A locally fine-tuned diffusion model for skateboard decals on **8 GB of VRAM**

KYLIAN ALGOET



CUSTOMER-DESIGNED DECKS, WITHOUT A DESIGNER FOR EVERY ONE-OFF ORDER

A user types a prompt, picks a trained style, optionally adds a reference image, and sees the artwork on a 3D deck.

13 MANDATORY REQUIREMENTS

One shapes everything: the model is **self-trained and runs locally, on consumer hardware**.

PRIMARY RESEARCH QUESTION

How can a locally fine-tuned diffusion model, conditioned on both a text prompt and a reference image, generate skateboard-decal artwork in multiple visually distinct styles with **reproducible quality** on consumer hardware with **8 GB of VRAM**?

12 SECONDARY QUESTIONS

8 ANSWERED WITHIN THEIR STATED SCOPE

4 BOUNDED • RQ4'S IMAGE-COUNT COMPONENT **INCONCLUSIVE**



PROJECT PLANNING

DeckForge AI - Project Planning

View 1

+ New view

Filter by keyword or by field

	Title	Assignees	Status
1	✓ M0 - Phase 0: research and repository foundation #1	KylianAlgoet	Done
2	✓ M1 - Prototype 0: static interactive 3D skateboard viewer #2	KylianAlgoet	Done
3	✓ M2 - Dataset research and dataset pipeline #3	KylianAlgoet	Done
4	✓ M3 - Prototype 1: local base-model benchmark #4	KylianAlgoet	Done
5	✓ M4 - Prototype 2: text and reference-image conditioning #5	KylianAlgoet	Done
6	✓ M5 - Prototype 3: local LoRA smoke test #6	KylianAlgoet	Done
7	✓ M6 - Prototype 4: style-learning experiments #7	KylianAlgoet	Done
8	✓ M7 - Prototype 5: integrated MVP #8	KylianAlgoet	Done
9	✓ M8 - Testing, deployment, and demo preparation #9	KylianAlgoet	Done
10	✓ M9 - Research report and PDF #10	KylianAlgoet	Done
11	✓ M10 - Presentation and jury preparation #11	KylianAlgoet	Done

PROJECT BOARD — ONE ITEM PER MILESTONE, WITH ITS STATUS.

M4 - Prototype 2: text and reference-image conditioning #5

Closed

KylianAlgoet/Selftrained-and-deployed-AI-image-generator

Public

KylianAlgoet opened 3w ago

Last edited by KylianAlgoet

Owner

Objective: Compare image-conditioning methods (RQ6, RQ7): img2img vs. IP-Adapter (ControlNet where relevant) on a fixed prompt + reference kit, with reference-strength sweeps and rubric evaluation.

Acceptance criteria:

- ✓ Fixed reference-image kit defined and frozen
- ✓ Methods compared on identical prompts/seeds with rubric scores
- ✓ Reference influence demonstrably controllable via strength parameter
- ✓ Strength sweep documented with output grids
- ✓ EXP rows complete

Planned start: 2026-08-04 | **Planned end:** 2026-08-06
Priority: critical
Dependencies: M3
Expected evidence: EXP records, comparison grids in docs/evidence/
Current status: Completed on 2026-08-01

ONE MILESTONE OPENED — OBJECTIVE, ACCEPTANCE CRITERIA, EVIDENCE, STATUS.

PUBLIC GITHUB PROJECT

Milestones M0–M12 · objectives · acceptance criteria · evidence · status

OPEN LIVE PLANNING ↗

<https://github.com/users/KylianAlgoet/projects/1/views/1>



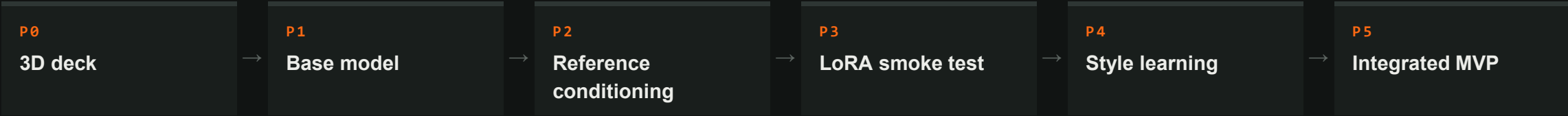
8 GB CHANGED EVERY DECISION

DEVICE TOTAL 8187.5 MiB 200 MiB SPARE • 2.4 % OF THE CARD



DEVICE USED = CUDA CONTEXT + RESERVED POOL + DISPLAY – THE ONLY FIGURE COMPARABLE WITH THE CARD'S TOTAL.
REFERENCE-CONDITIONED PATH: THE WORST CASE, AND THE PRODUCTION CEILING. RTX 4060 LAPTOP GPU • WINDOWS 11.

PROTOTYPE LADDER – EACH STEP ANSWERS THE QUESTION THE NEXT ONE DEPENDS ON. NONE WAS SKIPPED.



40
REGISTERED EXPERIMENTS

17
DECISION RECORDS

31
COMPLETED GPU GENERATIONS

Two human approval gates against a rubric defined before review — the first blinded, the second labelled.



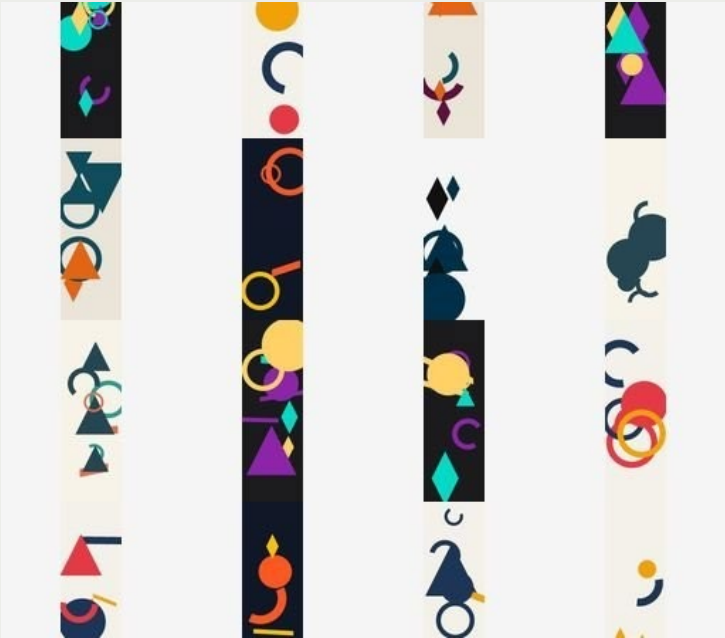
148 ITEMS. THREE STYLES. LICENSED BEFORE THEY WERE SPLIT.

EVERY ITEM CC0, PUBLIC DOMAIN, OR MADE FOR THIS PROJECT
LICENCE + SOURCE URL RECORDED **BEFORE** AN ITEM ENTERED A SPLIT



ukiyo-e 55

METROPOLITAN MUSEUM OF ART • CC0



minimal-geometric 52

GENERATED AND CURATED FOR THIS PROJECT



retro-poster 41

LIBRARY OF CONGRESS • PUBLIC DOMAIN

SPLIT 124 train • 17 validation • 7 holdout never in a training run.

CAPTIONS Style-only, chosen over verbatim descriptions by a blinded A/B comparison.



TRACK B — SDXL AT 1024 × 1024, ITS DESIGNED RESOLUTION.

SDXL WAS BETTER. IT WAS REJECTED ANYWAY.

SDXL scored **better** than SD 1.5 under the project's own rubric, at its designed resolution.

SD 2.1 could never be measured — authentication-gated.
The comparison rests on **two** candidates, not three.

10 738 MiB NEEDED

8187.5 MiB AVAILABLE

SPILLED TO HOST RAM

It did not crash. Windows spilled GPU memory into host RAM, so **all thirty benchmark runs completed and reported success.**

A zero exit code did not mean the model fit.

SD 1.5 SELECTED

Quality versus feasibility — and feasibility won.



REFERENCE



IMG2IMG OUTPUT



THE FREE METHOD LOST

dHash 0

Six outputs were flagged at perceptual-hash distance 6 or less. Two were **perceptually identical** to the reference. **Every flag was img2img. None were IP-Adapter.**

GEOMETRY	RUNS	MEDIAN dHASH	MIN
512 × 512	31	27	12
512 × 1536 PRODUCTION	9	5	0

LOWER = MORE SIMILAR. A REFERENCE ALREADY AT 512 × 1536 KEEPS 100 % OF ITS AREA, SO DENOISING AT STRENGTH 0.65 STARTS FROM AN ESSENTIALLY INTACT COPY.

IP-Adapter selected, scale 0.55. img2img costs zero extra VRAM and was rejected anyway — on a card with 200 MiB spare.

IMG2IMG AT THE DECK FORMAT, STRENGTH 0.65. TWO OF SIX FLAGGED OUTPUTS SCORED dHASH 0 — PERCEPTUALLY IDENTICAL TO THE REFERENCE.

TRAINING LONGER MADE IT LESS OBEDIENT

STEP 300 — SHIPPED

5

STYLE CONSISTENCY

4

PROMPT ADHERENCE

The model got better at the style by getting **worse at following the prompt** — the wrong trade for a product where the customer types what they want.

STEP 600 — REJECTED

5

STYLE CONSISTENCY

3

PROMPT ADHERENCE

Two adapters ship at step 300; **ukiyo-e is the exception and ships at 600**. Three per-style adapters, rank 8, default weight 0.7. Every run was checkpointed at 150 / 300 / 450 / 600 and scored.

WHAT DID NOT WORK



EXP-029 • RETRO-POSTER • STEP 300. DISPLAY TYPE READS “KTIRER”, “XETAXE”, “MAUNTIT REST”.

RETRO-POSTER — PARTIAL PASS

It learned the **frames, borders and display typography** of the archive posters, not only their style. It ships **labelled in the interface, on every request** — not dropped, and not re-scored until it passed.

RQ4 • IMAGE COUNT — INCONCLUSIVE

12 / 24 / 44 images, everything else fixed. The ordering came out non-monotonic with **one run per condition**, so an effect cannot be separated from variance. **No minimum image count is claimed.**

EVALUATION — THE BIGGEST LIMITATION

Visual scoring was AI-assisted: ChatGPT contributed visual analysis and proposed scores at the gates. I reviewed and approved every recorded score and kept final authority. **One approver, no second independent human rater — so no inter-rater agreement.**

NOT MEASURED

SD 2.1 was authentication-gated. Memorisation is **bounded, not settled** — 0 of 252 outputs flagged against the holdout, and dHash ≤ 6 is a coarse near-copy indicator, not proof.

**USER / BROWSER**`React + Three.js`**FASTAPI SERVICE**`one worker · busy lock · asserted in code at startup`**ONE RESIDENT PIPELINE**`Stable Diffusion 1.5 + per-style LoRA + optional IP-Adapter`**RTX 4060 LAPTOP GPU — 8 GB**`one generation at a time`

TWO PROCESSES. ONE GENERATION AT A TIME.

512 × 1536 DIRECT

Generated at the deck format, not square-then-crop. The hypothesis that the tall aspect would degrade quality was tested and refuted.

HTTP 409

A concurrent generation request gets a clean refusal, never an out-of-memory crash.

VALIDATED UPLOADS

Extension and MIME allowlist, real decode, size limits, random internal filenames — no user string reaches a filesystem path.

PROCEDURAL DECK

The 3D geometry is generated, so no third-party model licence enters the deliverable.



527

PYTEST

183

VITEST

38

PLAYWRIGHT END-TO-END

Clean clone

VALIDATED FROM SCRATCH

AUTOMATED TESTS PROVE BEHAVIOUR, NOT VISUAL QUALITY.

Only the human gates judged whether the pictures were good.

WHAT THE CLEAN-CLONE TEST IS FOR

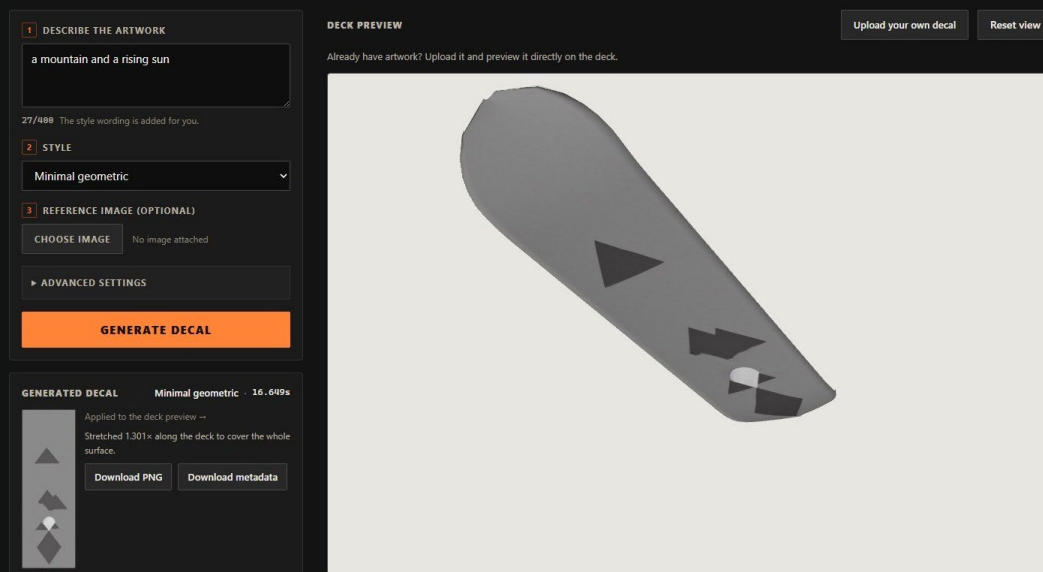
Clone into an empty directory, install from scratch, reach a working system.

It caught an integrity control that had only ever passed **on the machine that wrote it** — green for weeks while verifying nothing.

REPRODUCIBILITY HAS TWO HALVES

INFERENCE

Reproducible



CLEAN CLONE, FRESH ENVIRONMENT. SHA-256 IDENTICAL TO THE ORIGINAL RUN.

- Byte-identical output, three days later
- Freshly built environment, empty directory
- Verified by hash on every request

TRAINING

Not reproducible from the seed

- The LoRA initialisation draws from an **unseeded global RNG**
- The recorded seed never governed the part that mattered
- The three production checkpoints cannot be regenerated

The defect produced runs that looked completely valid. It is invisible unless you compare weights — and it was found only by **attempting** to reproduce a result instead of assuming it would.

The data pipeline is deterministic, and that was verified separately.



LOCAL, DELIBERATELY

FOUR OPTIONS COMPARED ON NINE CRITERIA

Native local NOT SELECTED	Docker NOT SELECTED GPU passthrough was never verified on this machine and its overhead is unmeasured — disqualifying on this margin.	Cloud NOT SELECTED Cost, plus a network dependency on the one day everything must work. The deciding reason: it answers nothing — the question is what fits in 8 GB of consumer hardware.	Native local + backup SELECTED Native Windows / NVIDIA, plus pre-generated backup demo assets.
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“No public URL — **by design** for this bachelor deployment.”

Internet hosting is not impossible. The measured constraint is narrower: **≈ 200 MiB worst spare prevents a second resident pipeline on the validated GPU**, so multi-worker GPU inference is unsupported on that machine.

≈ 200 MiB SPARE
ONE WORKER
OFFLINE BACKUP DEMO
CLEAN-CLONE PROOF

Reproducibility is carried by the runbook and the clean-clone proof, not by a public URL.



PRIMARY RESEARCH QUESTION – ANSWERED

Yes — within measured **bounds.**

SD 1.5 · 512 × 1536 · PER-STYLE LoRA AT 0.7 · IP-ADAPTER AT 0.55

2.4 %

Inference yes · training no

3 styles, 1 partial

SPARE VRAM UNDER REAL SERVING

REPRODUCIBILITY, SPLIT

RETRO-POSTER SHIPS LABELLED

WHAT IT DID
NOT PROVE

That LoRA beats the alternatives · that SD 1.5 is the best base model · that any image count is a threshold · that a green test suite means good pictures.



Prompt → **style** →
generation → **3D skateboard**

Local GPU. No cloud inference.



WHAT I WOULD CHANGE

REPEAT CONDITIONS

Repeat every condition before drawing a conclusion. The image-count question is inconclusive because each condition was run once — a **planning** mistake, not a resource one.

SEED AND VERIFY

Seed everything, then verify the **actual weights** rather than trusting the log. The training run looked valid the whole time.

SECOND RATER

Use a **second independent human rater**, so visual scoring can report agreement instead of one approver's judgement.

CONCLUSIONS STOP WHERE THE **EVIDENCE** STOPS.

THANK YOU

QUESTIONS